ModelIt!

Student Activity Pack

Microplastics in the Food Chain: The Invisible Invasion

Modeling Bioaccumulation and Trophic Transfer of Persistent Pollutants

Grade 09L2 | NGSS HS-LS2-6, HS-ESS3-3 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Where do you think the plastic from a water bottle ends up after you throw it away?**

2. **What are microplastics and how do they get into the ocean?**

3. **Draw a food chain and predict how a pollutant might move through it.**

4. **How might eating fish expose you to microplastics?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Plastic Input Rate | Microplastic Concentration | Bioaccumulation Factor | Trophic Transfer | Marine Organism Health | Decomposition Rate | Human Exposure Level

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Trajectory

Current input, zero decomposition, 50 years

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: 50 Percent Reduction

Half current input, existing stock persists

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Trophic Cascade

Track concentration through 4 trophic levels

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Chemical Leaching — Rate at which toxic additives like phthalates and BPA leach from plastic particles into organism tissue

Wastewater Treatment Efficiency — Percentage of microplastics removed during municipal water treatment before discharge

Bioremediation Potential — The capacity of engineered organisms or enzymes to break down microplastics in the environment

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a plastic bag thrown away in Kansas end up inside a fish you eat in California? Use evidence from your simulation to support your answers.

1 Why is microplastic accumulation essentially a one-way ratchet in your model?

2 How does biomagnification create a disproportionate burden on top predators and humans?

3 Why does reducing plastic input today not immediately solve the contamination problem?

4 What are the limitations of modeling microplastic fate with only 7 components?

5 How should society balance the benefits of plastic with its environmental persistence?

8 STEM Challenge: Design a Microplastic Monitoring and Reduction System

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YOUR ENGINEERING MISSION

Design a system that monitors microplastic levels across the food chain and identifies the most cost-effective intervention points.

THE SCENARIO

A coastal city's seafood industry is threatened by rising microplastic contamination. Design a monitoring system that tracks plastic from source to plate and a reduction strategy targeting the highest-impact intervention points.

Engineering Design Thinking

1. WHAT MONITORING TECHNOLOGIES WOULD YOU DEPLOY AT EACH STAGE OF THE FOOD CHAIN?
2. HOW WOULD YOU PRIORITIZE BETWEEN PREVENTING NEW PLASTIC INPUT VS. REMOVING EXISTING MICROPLASTICS?
3. HOW WOULD YOU COMMUNICATE RISK TO CONSUMERS WITHOUT DESTROYING THE FISHING INDUSTRY?
4. WHAT DATA WOULD PROVE YOUR INTERVENTION IS WORKING GIVEN THE DECADES-LONG TIMESCALE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental Toxicologists study how pollutants move through ecosystems and affect organism health, earning $60,000-$115,000/year at EPA, consulting firms, and universities.